

Nicholas Lee *Data Scientist*

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Profile

Results-driven Data Scientist with expertise at the intersection of advanced statistical analysis, scalable data engineering, and product development.

Lead data scientist on a cross-functional team, designing and managing end-to-end ETL pipelines that transform raw consumer transaction data into actionable insights.

Skilled in feature engineering, missing data imputation, bias correction, and predictive modeling to forecast consumer behavior at scale.

Hold a PhD in Astronomy with a strong foundation in physics, mathematics, statistics, and machine learning.

Passionate about leveraging data to solve complex problems and drive data-informed decision-making.

Skills

Languages

Python, SQL, JavaScript, Java

Frameworks & Libraries

Tensorflow, PyTorch, XGBoost, Hugging Face, Scikit-learn, spaCy, NLTK

ML & Data Tools

Spark, Databricks, MLflow, Airflow, Kubeflow, AWS Glue, Snowflake, BigQuery

Infra & Cloud

Kubernetes, Docker, AWS, Azure, GCP, Terraform, Jenkins, GitLab CI, SageMaker, ECS, Lambda

Experimentation & Evaluation

A/B testing, model calibration, causal inference, statistical testing frameworks

MLOps & Deployment

CI/CD pipelines, real-time inference systems, IaC

Professional Experience

Lead Data Scientist

Bloomberg LP

01/2021 – Present
San Francisco, CA, US

- Led a cross-functional team of data scientists, data engineers, and analysts, delivering high-quality data products that leveraged alternative data sources (consumer transactions, geospatial data, web activity) to generate insights on consumer behavior and company performance.

- Architected and scaled ETL pipelines using Python (pandas, PySpark), SQL, and Airflow, ensuring reliable ingestion, transformation, and enrichment of billions of records across multiple data domains.
- Applied advanced statistical modeling, machine learning (scikit-learn, XGBoost, TensorFlow), and natural language processing (spaCy, Hugging Face Transformers) to identify key behavioral signals, impute missing data, and forecast business outcomes with >90% accuracy.
- Partnered with data engineering teams to design scalable data infrastructure in AWS (S3, Redshift, EMR, Glue) and Snowflake, optimizing performance and reducing pipeline costs by 20%.
- Directed the adoption of MLOps best practices by integrating MLflow and Docker/Kubernetes, enabling reproducible experiments, versioned model deployments, and continuous monitoring of data/model drift.
- Championed a collaborative, agile workflow, leading sprint planning, code reviews, and stakeholder alignment with product managers and business leaders to ensure deliverables were aligned with strategic priorities.
- Mentored junior team members in statistical methods, feature engineering, data visualization (Tableau, Power BI), and communication of technical findings to non-technical stakeholders, improving team productivity and delivery speed.
- Pioneered initiatives in Generative AI, developing LLM-powered prototypes using OpenAI GPT, LangChain, and Hugging Face to automate insights extraction and accelerate data-driven decision-making.
- Integrated LLM-based agents into internal data workflows to enhance query automation, knowledge retrieval, and client reporting, reducing manual analysis time by 40%.

Product Data Scientist

Second Measure

09/2017 – 01/2021
San Francisco Bay Area

- Partnered with investors and business stakeholders to translate consumer purchase data into actionable insights, enabling smarter investment decisions across retail, fintech, and consumer goods sectors.
- Designed and maintained data pipelines with SQL, Python (pandas, NumPy), and Spark, processing billions of anonymized credit card transactions into structured datasets optimized for financial analysis.
- Built predictive models using scikit-learn, XGBoost, and time-series forecasting techniques to anticipate company performance trends and consumer behavior patterns, reducing client decision cycles by 30%.
- Created data visualizations and dashboards in Tableau and Looker, highlighting KPIs such as customer acquisition, churn, market share shifts, and revenue trajectories to non-technical audiences.
- Collaborated closely with product managers and engineers to define data requirements, design new data features, and integrate insights into customer-facing analytics products.
- Applied statistical methods (A/B testing, regression, survival analysis) to validate consumer behavior hypotheses and quantify confidence in investment theses.

- Drove improvements in data quality and governance, implementing anomaly detection and validation checks to ensure accurate, trustworthy datasets for institutional clients.
- Presented findings to executive-level investors and hedge fund clients, translating technical analyses into clear narratives that influenced multi-million-dollar investment strategies.

Insight Data Science Fellow

06/2017 – 08/2017

Palo Alto, CA, US

Insight Data Science

- Developed 7 data stories for Joy's inaugural data blog, increasing web traffic and brand recognition through data-driven storytelling.
- Analyzed data from 50,000+ weddings to provide insights that guided market research and informed product development.
- Predicted user gender from first names by comparing against the US Social Security Administration baby names database and applying a Naive Bayes Classifier.
- Assessed the impact of guest travel distance on wedding attendance across 5,000+ weddings by converting unstructured text inputs into geographic coordinates using text parsing, ZIP code lookups, and the OpenStreetMap API.
- Applied k-Means clustering to segment user demographics, identifying regions where marketing strategies needed adjustment to better match local user behavior.

Post-Doc

09/2015 – 05/2017

Niels Bohr Institute, University of Copenhagen

- Conducted research on galaxy formation and evolution, focusing on star-formation processes and the mechanisms driving galactic quenching ("how galaxies die").
- Led a team of scientists in developing a data tool to query, collect, and visualize astronomical datasets from large-scale online repositories.
- Designed and taught a 10-week seminar on Python and Clean Coding practices, establishing a baseline coding proficiency and enforcing consistent coding standards across the team.
- Introduced modern software engineering practices—including agile methodology, Scrum, Git workflows, pair programming, code reviews, and unit testing—into an academic research setting, significantly improving collaboration and code reliability.

Data Scientist

11/2014 – 06/2015

Honolulu, HI, US

Diagenetix, Inc.

- Headed a new data science team to develop predictive models of fungal growth in grape crops, improving efficiency of fungicide application in large-scale agriculture.
- Increased model accuracy from 50% to 80% by implementing a logistic regression algorithm trained on weather features such as temperature, humidity, and rainfall.

- Simulated fungal spread via wind patterns, demonstrating predictive capabilities with enhanced environmental data integration.
- Optimized data collection frequency by training linear regression models on simulated datasets, balancing measurement costs with algorithm accuracy.
- Presented model insights to the CEO, illustrating how predictive analytics could create substantial product value and save customers millions in crop protection costs.

Graduate Student

08/2008 – 06/2015

Hawaiian Islands

Institute for Astronomy, University of Hawaii

- Characterized 100,000+ galaxies to study star formation, galaxy growth, and quenching, classifying key lifecycle stages and statistically identifying outliers.
- Applied Bayesian linear regression to measure physical processes governing star formation, accounting for uncertainty across multiple variables.
- Published 25 peer-reviewed articles (including 4 first-author publications) and presented research at 15 international conferences, establishing expertise in astrophysics and statistical modeling.
- Developed leadership and teaching skills through workshops on leadership, diversity, equity, and professional development, while leading multiple teaching teams.
- Taught courses in astronomy, statistics, professional and career development, education, and programming, mentoring students across technical and professional domains.

Inquiry Lab Instructor

03/2012 – 09/2014

Institute for Scientist & Engineer Educators/Akamai Workforce Initiative

- Participated in the Professional Development Program (ISEE/AWI, 2012–2013), gaining training in inquiry-based teaching techniques.
- Designed, facilitated, and evaluated an inquiry-based lab on Gaussian statistics for six undergraduate summer research fellows at UH Mānoa (2012).
- Led a team of four instructors to design, facilitate, and evaluate an inquiry-based lab on the properties of light, teaching 20 community college students in a STEM bridge program at Kapiʻolani Community College (2013).
- Developed instructional strategies to improve student engagement, critical thinking, and applied problem-solving in STEM education.

Astronomy 110 Summer Instructor

06/2012 – 08/2012

University of Hawaii at Manoa

- Co-taught Astronomy 110 (Astronomy for Non-Majors) to a summer session of 14 students, implementing active learning methods such as lecture tutorials and Think-Pair-Share questions.
- Designed course materials including lesson plans, syllabus, lectures, homework assignments, and exams.
- Lectured and facilitated classes, labs, and group work, fostering engagement and collaborative learning.

Education

Doctor of Philosophy, Astronomy <i>University of Hawaii at Manoa</i>	2010 – 2015
Master's degree, Astronomy <i>University of Hawaii at Manoa</i>	2008 – 2010
Bachelor of Science, Astrophysics and Physics <i>University of California, Berkeley</i>	2003 – 2007
Bachelor's degree, Physics & Astronomy <i>University of California, Berkeley</i>	2003 – 2007